C4 falsifier: exponent-space structure of decompositions on PKM's Kummer tower

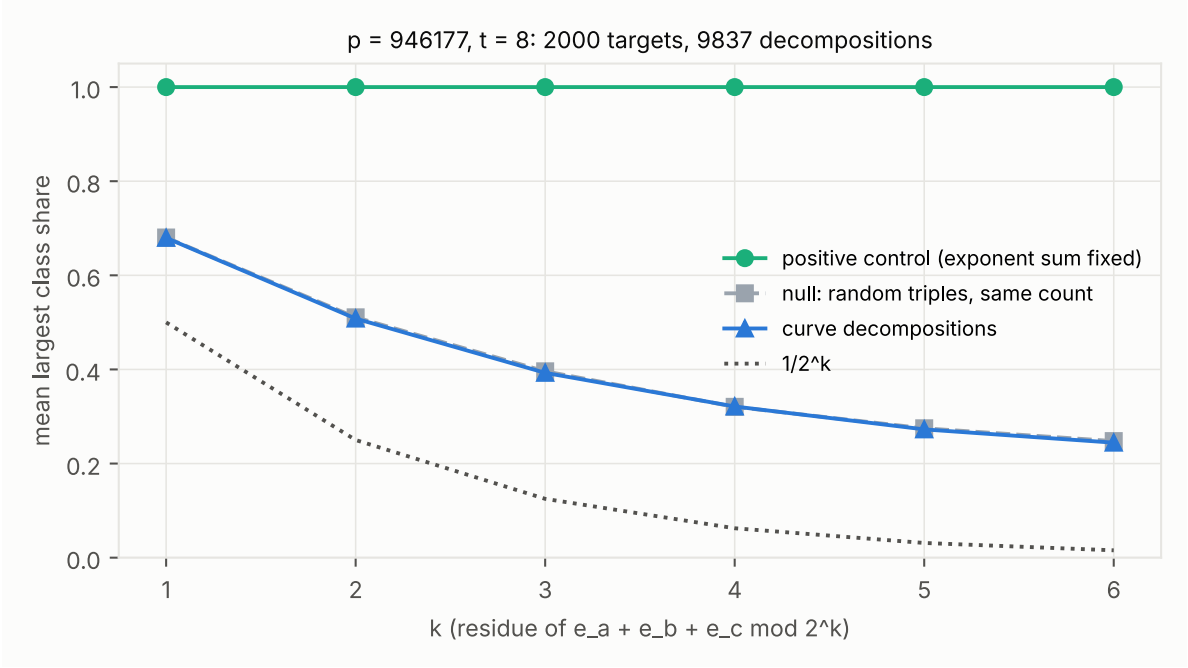
2026-10-13. `research/ic_exponent_falsifiers_20261013/`, program `examples/kummer_exponent_bias.rs`. **Class: falsification, negative.**

Question

On PKM's tower $V = \mu_{2^t}$ every base abscissa is ζ^e. A representation-technique search for $R = P_a + P_b + P_c$ needs a linear form of the exponents that the target predicts. Simplest form: is $e_a + e_b + e_c \bmod 2^k$ concentrated among the decompositions of one target?

Method

Curve `icv1-fp20-tm728-d215a2a1` ($p = 946177$), $t = 8$, 246 signed base points. 2,000 uniform targets; every decomposition enumerated exactly and counted once (9,837 in all). Statistic: mean over targets of the largest residue-class share of the exponent sum mod 2^k. Null: random exponent triples, same count per target. Positive control: the same detector on F_p^* multiplication, where the sum is fixed by the target.



Largest class share per k: the curve lies on the null; the positive control reads 1.

k	$1/2^k$	curve	null	positive control
1	0.500	0.680	0.680	1.000
2	0.250	0.508	0.511	1.000
3	0.125	0.393	0.396	1.000
4	0.0625	0.321	0.321	1.000
5	0.0313	0.273	0.275	1.000
6	0.0156	0.244	0.248	1.000

Decision

The exponent sum of a decomposition carries no information about the target (curve and null differ by under 1 % at every k). C4′s simplest form is falsified on this base, as predicted: the elliptic group law does not respect the multiplicative structure of μ_N. Scope: one curve, one tower height, the plain sum.

Environment: Claude Code cloud container, Intel Xeon @ 2.10 GHz, 4 vCPU, rustc 1.97.0. Conductor control plane unreachable: no task card.